NAME

mbgrd2octree − Translates a topography grid to a TRN octree topography model for use with the MB-System TRN (terrain relative navigation) tools.

VERSION

Version 5.0

SYNOPSIS

mbgrd2octree **−−input=***inputGridFile* **−−output=***outputOctreeFile* [**−−bounds=***val1/val2/val3/val4* **−−verbose −−help**]

DESCRIPTION

Mbgrd2octree converts a topography grid into a TRN (terrain relative navigation) octree topography model file for use with the **MB-System** TRN tools (e.g. **mbtrnpp**). The input grid is expected to be a GMT/netCDF grid in a cartesian projected coordinate system such as UTM, with dimension variables named "x" and "y" (or, failing that, "lon" and "lat") and a data variable named "z", each carrying an "actual_range" attribute from which the grid node coordinates are reconstructed. **Mbgrd2octree** derives from **grdOctreeMaker**, a program contained within the **terrain-nav** codebase developed over many years by Steve Rock's Precision Navigation project at Stanford.

The program reads the input grid, reconstructs the north (or latitude) and east (or longitude) coordinate vectors from the "actual_range" attributes, and treats each node's grid value as a positive-down depth in a NED (north, east, down) coordinate frame, consistent with the octree format's historical convention. Grid nodes whose value equals the no-data sentinel 99999, or that are NaN, are skipped. Only nodes with a depth between 1 and 3500 meters are accepted; these depth limits, and the base horizontal/vertical octree resolution (1 in the grid's native units), are presently fixed at compile time and are not exposed as command line options.

Mbgrd2octree automatically determines the horizontal extent of the octree from the range of accepted grid nodes, and automatically derives the octree resolution from the coordinate spacing between the first two grid nodes on each axis. An octree object is constructed with dimensions that are a power-of-two multiple of this resolution and that enclose the full point cloud. Each accepted surface point is added to the octree, and then two additional voxels beneath each surface point are filled (using the octree's actual/effective vertical resolution) so that steep slopes do not present holes when the resulting model is viewed obliquely. The octree is then compressed (collapsed) and written to the output file.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−help

This "help" flag cause the program to print out a description of its operation and then exit immediately.

−−input=*inputGridFile*
Sets the filename of the input topography grid. This must be readable as a netCDF grid with "x"/"y" (or "lon"/"lat") dimension variables and a "z" data variable. If the supplied name does not already end in ".grd", the suffix ".grd" is appended to it. This option (or the equivalent short option **−I**) must be specified.

−−output=*outputOctreeFile*
Sets the filename of the output TRN octree file. If the supplied name does not already end in ".bo", the suffix ".bo" is appended to it. This option (or the equivalent short option **−O**) must be specified.

−−bounds=*val1/val2/val3/val4***, −R***val1/val2/val3/val4*
Sets an optional bounding filter on the grid nodes used to build the octree, given as four slash−separated numeric values that must satisfy *val1* < *val2* and *val3* < *val4*. Grid nodes are excluded whenever their north coordinate is less than *val3* or greater than *val4*, or whenever their east coordinate is greater than *val1* or less than *val2*. Setting *val1* or *val2* to −1 disables that particular east/west test. Because *val1* must be less than *val2*, supplying two finite (non−−1) values for both normally causes every node to be excluded on the east/west axis, so in practice at most one of the two east/west values should be set to something other than −1. If **−−bounds**/**−R** is omitted altogether, no additional bounding filter is applied beyond the fixed depth−range and no−data checks described above.

−−verbose
The **−−verbose** option causes **mbgrd2octree** to print out status and debug messages; repeating it (e.g. as **−V −V**) redirects this output to stderr and increases the level of detail reported.

SEE ALSO

mbsystem(1), **mbgrdtilemaker**(1), **mbtrnpp**(1), **mbgrid**(1)

BUGS

Maybe. Maybe not.